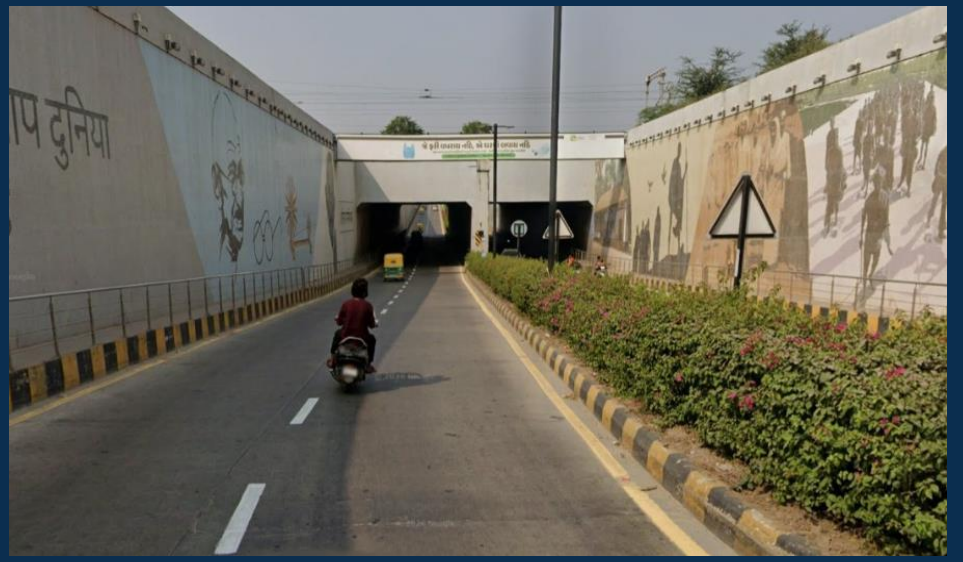
Flood-Resilient Emergency Transit Plan for Gandhinagar

Corridor: Mahatma Mandir Underpass stretch with access to Civil Hospital Gandhinagar

Team: KAIR for Cities

Members: Ashwini, Kaira, Radhe, Ishani

Problem Statement 1 · Track 1: Urban Flash-Flooding & Emergency Corridor Preservation



Google street view



https://www.divyabhaskar.co.in/local/gujarat/gandhinagar/news/water-filled-in-vavol-mahatma-mandir-underpass-temperature-26-degrees-134993062.html

Why Flooding Cuts Lifelines in Gandhinagar?

Heavy rainfall & waterlogging
Intense monsoon rainfall can rapidly increase water accumulation in low-lying stretches

Blocked underpasses & arterial roads
Sectors get cut off from emergency healthcare

Ambulance & emergency delays
Vehicles stall without real-time depth information

Inefficient stormwater drainage
Urban debris chokes drains and gratings

Paved surfaces
Paved surfaces reduce natural infiltration and increase surface runoff, contributing to urban waterlogging.

KEY PROBLEM

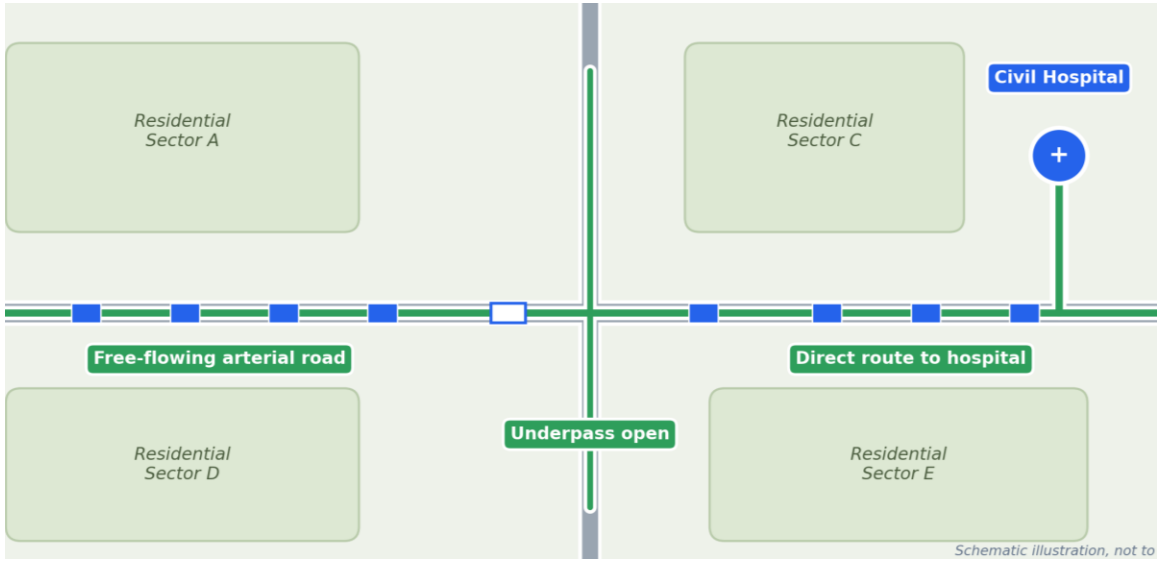
How can we maintain emergency mobility while reducing urban waterlogging during extreme rainfall?

Vulnerability Mapping: Where the Corridor Floods



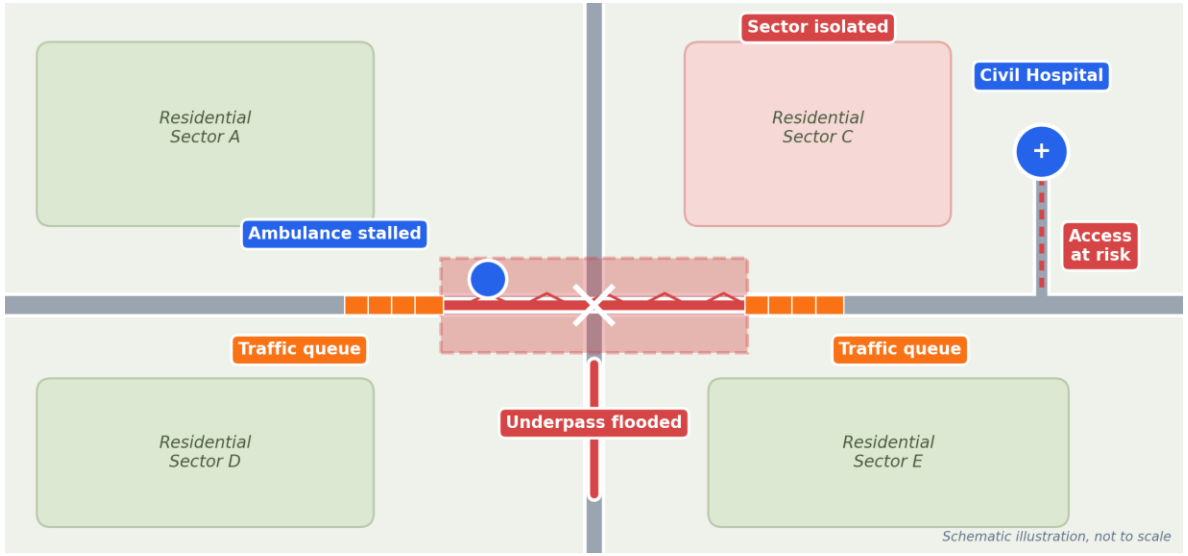
What Happens When the Corridor Floods

Normal day



- Free-flowing traffic on arterial roads
- Underpasses open to ambulances and buses
- Sectors connected to Civil Hospital

During a cloudburst



- Water pools at underpasses and low points
- Roads inaccessible; hospital access at risk
- Congestion and isolated residential sectors

Emergency Route Continuity Plan



Life-Line Lanes
Elevated, priority lanes reserved for ambulances, rescue teams and buses

Hospital access
Pre-marked safe routes to Civil Hospital from every sector

Pedestrian evacuation
Signed walking routes to higher, dry ground

Low-Cost Urban Sponge Interventions

Rain gardens

What: Planted depressions that hold runoff.

Where: Roadside verges near underpass approaches.

How it helps: Slows and soaks water before it reaches the road.

Bioswales

What: Shallow vegetated channels.

Where: Road edges, leading to catchments.

How it helps: Carries and filters overflow away from low points.

Permeable paving

What: Porous paving on road shoulders.

Where: Shoulders and bus bays near underpasses.

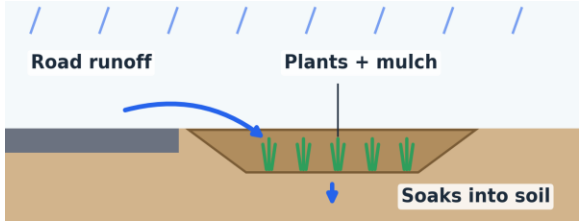
How it helps: Lets rain percolate instead of pooling.

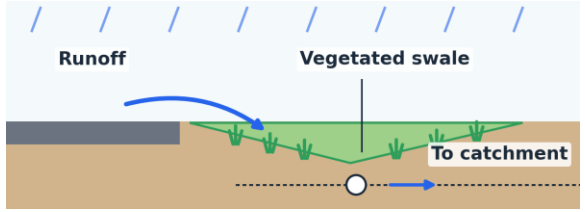
Soak pits & drains

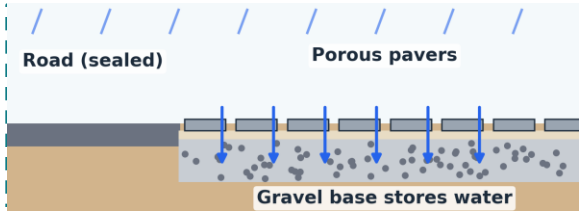
What: Gravel-filled pits and desilted drains.

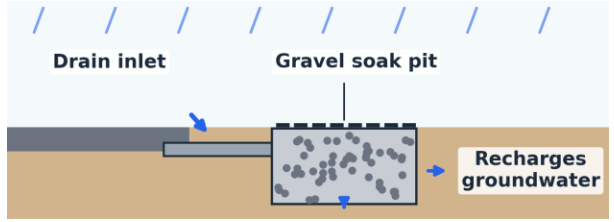
Where: Low spots and drain inlets.

How it helps: Recharges groundwater and clears choked flow.



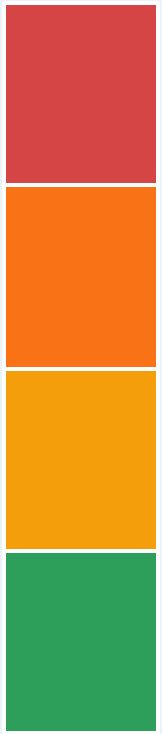






Early Warning & Hazard Communication

Underpass depth gauge (concept)



STOP: above 50 cm

Danger: 30–50 cm

Caution: 10–30 cm

Safe: below 10 cm

Indicative thresholds; finalise with municipal engineers.

Warning barriers

Boom barriers and flashing lights close the underpass when depth crosses the red level.

Digital hazard boards

Boards at approach junctions show live status: OPEN, CAUTION or CLOSED.

WhatsApp / SMS alerts

Ward-level groups push rainfall and closure alerts to residents and drivers.

Alternative route info

Signs and board messages point vehicles to the nearest Life-Line Lane.

First 6 Hours: Who Does What

Time	Ward officials	Traffic police	Emergency teams	Volunteers
0–2 hours	Monitor rainfall; inspect and clear drains	Station staff at underpass entries; check barriers	Pre-position ambulances near Civil Hospital	Clear debris from gratings; share alerts
2–4 hours	Deploy pumps; close unsafe roads	Activate diversions; keep Life-Line Lanes clear	Prioritise ambulance routing	Guide traffic; update hazard boards
4–6 hours	Restore critical connectivity	Manage congestion; coordinate reopening	Rescue and evacuation support	Assist pedestrian evacuation

Implementation & Feasibility

1. Immediate

- Signboards and depth markers
- Warning barriers
- Drain and grating cleaning

2. Short-term

- Rain gardens and bioswales
- Soak pits
- Permeable shoulders

3. Long-term

- Improved drainage capacity
- Permanent emergency corridor planning

Materials & cost heads

Signs, barriers, plants, gravel, porous paving. Costs: materials, labour, desilting, upkeep.

Responsible departments

Gandhinagar Municipal Corporation, R&B Dept., Traffic Police, 108 Emergency Services

Maintenance

Pre-monsoon desilting, seasonal plant care, regular checks of barriers and boards

Expected Impact

- **Faster emergency vehicle access**
- **Less water on the road**
- **Quicker public hazard awareness**
- **Safer pedestrian evacuation**
- **Better stormwater management**

CONCLUSION

“A flood-resilient city is not just one that prevents flooding, but one that keeps essential services moving even during floods.”